

JAVA SDE-2 INTERVIEW PREPARATION SERIES

FRAMEWORKS, DATA, AND MESSAGING - BOOK 02 OF 12 - STUDY STEP FW-02

Hibernate and JPA for Java Backend Interviews

Entity Lifecycles, Mapping, Fetching, Transactions, and Query Discipline

BY

Vinay Reddy Kalluri

A SQL-first, executable guide to JPA contracts, Hibernate lifecycle, associations, fetching, batching, locks, caches, diagnostics, and live SDE-2 interviews.

JPA | HIBERNATE | ENTITIES | FETCHING | LOCKING

JAVA 21 | INTERVIEW CORE | SDE-2 FOLLOW-UPS | PRINTABLE EDITION

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Open educational interview-preparation edition

Start Here - Your Route Through This Volume

60-second entry rule: If two or more recognition signals below expose a gap, begin with Chapter 1. If the prerequisites are weak, step back to FW 01 - MySQL. If you can already explain every outcome without notes, jump to Part II and prove it through the practice ladder.

Teach ORM behavior only after relational and transaction fundamentals, then make persistence-context identity, flush, fetching, query shape, batching, locking, caching, equality, and diagnosis observable and testable.

Readiness map

Decision	Evidence
START HERE	Object state must be persisted relationally; Lazy loading or N+1 queries affect behavior; Transactions and entity lifecycle interact.
PREPARE FIRST	FW-01 MySQL; Java object and exception fundamentals.
MOVE ON WHEN	Separate JPA specification contracts from Hibernate provider behavior; Reason about entity lifecycle, identity, dirty checking, flush, ownership, cascades, and fetching; Prevent N+1, stale bulk state, equality, batching, locking, cache, and transaction-boundary failures.

Choose the route that fits today

- 1 **Learning:** read in order, type the representative Java examples, and complete each dry run.
- 2 **Revision or rehearsal:** start at the first decision map, invariant, or failure mode you cannot explain; if none expose a gap, go directly to Part II and prove readiness without notes.

Frameworks Segment Position

FW 01 - MySQL	YOU ARE HERE FW 02 - Hibernate and JPA	FW 03 - Spring Framework
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A note from Vinay

An ORM does not remove SQL or transaction reasoning; it schedules it. Keep the entity state, persistence context, flush point, and generated SQL visible in the same mental model.

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Part I - Core Learning**SDE-2 CORE CHAPTER**

Chapter 1 - Hibernate and JPA for Java Backend Interviews

Learning Path: SQL Before ORM

Jakarta Persistence defines portable object-persistence contracts. Hibernate ORM implements those contracts and adds provider-specific capabilities. Neither changes the fundamental database work: rows are read, constraints arbitrate races, transactions commit, and SQL plans consume resources.

From Vinay: Before approving an entity mapping, write the SQL you expect it to produce. If you cannot predict the query count, join shape, lock scope, and commit point, the annotation has hidden too much. This book makes that hidden work visible one layer at a time.

Prerequisite bridge from MySQL

Given:

SQL

```
CREATE TABLE purchase_order (
  order_id BIGINT PRIMARY KEY,
  customer_id BIGINT NOT NULL,
  status VARCHAR(20) NOT NULL,
  total_cents BIGINT NOT NULL,
  version BIGINT NOT NULL,
  created_at TIMESTAMP NOT NULL
);
```

JDBC can issue an explicit optimistic update:

SQL

```
UPDATE purchase_order
SET status = ?, version = version + 1
WHERE order_id = ? AND version = ?;
```

JPA expresses the same intent with a managed entity and `@Version`; Hibernate eventually generates a version-checked `UPDATE`. The abstraction removes repetitive mapping, not the concurrency requirement.

Dependency order

TEXT

SQL + JDBC + transaction boundary

- > JPA contract and Hibernate implementation
- > entity identity and mapping
- > persistence context and lifecycle
- > dirty checking and flush
- > associations and ownership
- > proxies, fetching, and N+1
- > JPQL/Criteria/native/projection/pagination
- > batching, locks, caches, bulk work
- > tests and production diagnosis

The SQL prediction card

Before a repository or `EntityManager` call, answer:

- 1 Does this call execute SQL now, or only change persistence-context state?
- 2 Which tables, columns, and predicates will appear?
- 3 How many statements can this object traversal trigger?
- 4 When will a constraint or optimistic conflict be observed?
- 5 Which transaction and connection own the work?
- 6 Is returned state managed, detached, projected, or proxied?

Contract boundaries

Layer	Owns	Does not guarantee
JPA specification	portable lifecycle, mapping, query, locking contracts	Hibernate-specific SQL shape or every optimization
Hibernate	implementation, dialect, batching, fetch/caching extensions	database plan or network reliability
JDBC driver/pool	protocol, binding, connections, timeout mechanisms	ORM graph correctness
database	constraints, SQL execution, isolation/durability per configuration	application idempotency or external side effects

First failure timeline

TEXT

```
find entity -> mutate field -> call remote API -> flush -> constraint fails
```

The remote side effect already happened even though the database transaction rolls back. Place remote work outside the transaction or use an outbox/state-machine design. Calling `save()` earlier does not make two resource managers atomic.

Quick check

- 1 Which part of optimistic locking is still performed by SQL?
- 2 Why can a field assignment appear successful before a constraint failure?
- 3 What does JPA standardize that Hibernate implements?

Practice

- **Foundation:** Translate an entity lookup and update into expected SQL.
- **Interview Core:** Mark which operations need a transaction and which merely read.
- **SDE-2 Follow-up:** Redesign the failure timeline so the external effect is recoverable.

Continue when you can explain the difference between Java state change, SQL flush, and transaction commit.

SDE-2 CORE CHAPTER

Chapter 2 - JPA Contract, Hibernate Provider, and Bootstrap

The names matter

- **Jakarta Persistence (JPA)** is the specification and API: annotations such as `@Entity`, `EntityManager`, JPQL, lifecycle, and portable locking rules.
- **Hibernate ORM** is a provider that implements JPA and offers native APIs/extensions.
- **Spring Data JPA** builds repository abstractions over JPA; it is covered in the next volume.

Saying "JPA is an ORM" is imprecise. It is the contract used by ORM providers.

Minimal entity

JAVA

```
@Entity
@Table(name = "purchase_order")
public class PurchaseOrder {
    @Id
    @Column(name = "order_id")
    private Long id;

    @Column(nullable = false, length = 20)
    private String status;

    protected PurchaseOrder() {}

    public PurchaseOrder(Long id, String status) {
        this.id = Objects.requireNonNull(id);
        this.status = Objects.requireNonNull(status);
    }
}
```

An entity needs a no-argument constructor that is public or protected under the portable contract. Avoid final entity classes and final persistent accessors when proxying/lazy behavior may require subclassing.

Java SE lifecycle

JAVA

```
EntityManagerFactory factory =
    Persistence.createEntityManagerFactory("orders");

try (EntityManager entityManager = factory.createEntityManager()) {
    EntityTransaction transaction = entityManager.getTransaction();
    transaction.begin();
    try {
        entityManager.persist(new PurchaseOrder(1L, "CREATED"));
        transaction.commit();
    } catch (RuntimeException failure) {
        if (transaction.isActive()) {
            transaction.rollback();
        }
        throw failure;
    }
}
factory.close();
```

EntityManagerFactory is expensive and normally application-scoped. **EntityManager** represents a persistence-context/unit-of-work boundary and is not shared concurrently across requests.

Spring usually manages these resources and transactions, but the underlying scope still matters.

Configuration and schema ownership

`persistence.xml`, programmatic configuration, or framework properties define provider, datasource, mappings, dialect, schema validation, logging, batching, and cache behavior.

For production, prefer versioned database migrations. ORM schema generation is useful for disposable tests and prototypes. A startup **update** strategy is not a substitute for reviewed, reversible migrations against production-scale data.

Exceptions and transaction state

Persistence failures often surface at flush or commit. After a runtime persistence exception, assume the current transaction must roll back. Do not catch a constraint exception and continue to write unrelated managed entities in the same unit of work.

Exception types may be JPA-standard (**OptimisticLockException**, **EntityNotFoundException**, **TransactionRequiredException**) or provider/driver-specific beneath them. Translate at a stable application boundary while preserving diagnostic cause and constraint identity.

When direct JDBC is clearer

Use SQL/JDBC for bulk updates, highly tuned reporting, database-specific features, or hot paths where entity tracking adds no value. Use JPA for domain-oriented transactional changes where identity, associations, dirty checking, and optimistic locking reduce repeated mapping. Mixed use requires awareness that direct SQL can make the persistence context stale.

Debug exercise

Broken: create one static `EntityManager` and use it from every request.

Why: it is not a thread-safe global cache; it mixes units of work, grows managed state, and creates concurrency hazards.

Repair: inject/create a transaction-scoped manager per unit of work and keep the factory/pool shared.

Quick check and practice

- 1 Who generates the SQL: the JPA specification or provider?
 - 2 Which object is expensive and shared, and which is unit-of-work scoped?
 - 3 Why should production DDL use a migration system?
- **Foundation:** Bootstrap the lab and persist one entity.
 - **Interview Core:** Compare direct JDBC with a managed entity update.
 - **SDE-2 Follow-up:** Define an exception translation policy that preserves retry classification.

SDE-2 CORE CHAPTER

Chapter 3 - Mapping Identity, Values, and Schema Boundaries

An entity mapping is executable database design. The annotations should match-not invent-the table's identity, nullability, width, uniqueness, and relationships.

Access type

Place `@Id` on a field for field access or on a getter for property access. Mixing annotations without explicit `@Access` creates confusing behavior. With field access, Hibernate reads/writes persistent fields directly; domain methods can still protect mutation.

Identifier strategies

Common choices include assigned IDs, database identity columns, sequences where supported, and UUIDs. Compare:

- ability to know ID before insert;
- insert batching behavior;
- database portability;
- allocation/gaps;
- key width and locality;
- cross-service generation.

Do not implement `equals` around a generated ID as though two transient entities with null IDs are equal.

Value types and embeddables

JAVA

```
@Embeddable
public class Money {
    @Column(name = "amount_cents", nullable = false)
    private long amountCents;

    @Column(name = "currency", nullable = false, length = 3)
    private String currency;
}
```

An embeddable has value semantics and no independent entity identity. Keep invariants in its constructor/methods. A Java `record` can be useful in other layers; verify provider support and persistence requirements before treating every value as an entity.

Enums and converters

Prefer `@Enumerated(EnumType.STRING)` over ordinal storage because reordering constants changes ordinal meaning. Renaming a string constant is still a data migration.

An `AttributeConverter` maps a domain value to one column. Keep conversion deterministic and side-effect free. It does not replace a database constraint and may affect query parameter conversion.

Inheritance is a database trade-off

- `SINGLE_TABLE`: one table, discriminator, fast polymorphic query, nullable subtype columns.
- `JOINED`: normalized subtype tables, joins for polymorphic reads.
- `TABLE_PER_CLASS`: duplicated columns and union-like polymorphic access.

Often composition or separate aggregates are clearer. Choose inheritance only for a true subtype model and inspect generated SQL.

Identity and equality

A robust equality strategy must survive transient, managed, proxied, and detached states.

Options:

- immutable, database-unique business key used from construction;
- generated identifier equality only after assignment, with careful collection use;
- reference identity within a persistence context when value equality is unnecessary.

Never include mutable fields or associations in `hashCode` for an entity placed in a `HashSet`; changing them makes the object unreachable in its bucket. Hibernate proxies also make strict `getClass()` comparisons tricky; use a provider-aware strategy if entities can be proxied.

Schema constraints remain authoritative

JAVA

```
@Column(nullable = false, length = 20)
private String status;
```

This helps metadata and generated schema; it is not proof the production table matches. Keep migrations, database constraints, Bean Validation, and domain validation aligned. Each fails at a different boundary.

WATCH OUT | Mapping edge cases

Mapping	Failure	Better reasoning
double money	approximation	exact minor units/decimal value
ordinal enum	reorder corrupts meaning	stable code/string plus migration plan
mutable natural ID in hash	broken set/map membership	immutable unique key
@Lob for ordinary text	fetch/storage surprises	choose DB type from size/access
entity for every table	domain graph mirrors schema accidentally	model aggregate behavior deliberately
eager relationships everywhere	joins/query storms	per-use-case fetch plan

Quick check and practice

- 1 How is entity identity different from value equality?
 - 2 Why is `EnumType.STRING` safer but not migration-free?
 - 3 What can schema annotations not prove about production?
- **Foundation:** Map an immutable `Address` embeddable.
 - **Interview Core:** Choose ID generation for a high-write order table and justify it.
 - **SDE-2 Follow-up:** Design equality for a proxied entity with an immutable business key.

SDE-2 CORE CHAPTER

Chapter 4 - Lifecycle, Persistence Context, Dirty Checking, and Flush

Four lifecycle states

TEXT

```
new/transient --persist--> managed --remove--> removed
                        |
                        +--detach/clear/close--> detached
detached --merge--> state copied into a managed instance
```

- **New:** ordinary object with no persistent identity in this context.
- **Managed:** tracked; changes may be synchronized automatically.
- **Detached:** has persistent identity but is no longer tracked.
- **Removed:** scheduled for deletion while managed.

Identity map and repeatable Java identity

Within one persistence context, a given entity type and database identity correspond to one managed instance. Repeated `find` calls can return the same Java object without another SQL query.

This first-level cache is mandatory context behavior, not a general cross-request cache. It can also become stale when bulk SQL or external writers change the row.

Dirty checking

JAVA

```
PurchaseOrder order = entityManager.find(PurchaseOrder.class, orderId);
order.markPaid();
// no persist/save call required for an already-managed entity
```

Hibernate tracks managed state and detects changes. At flush it can generate:

SQL

```
update purchase_order
set status=?, total_cents=?, version=?
where order_id=? and version=?
```


The exact columns depend on mappings/provider features such as dynamic update. Predicting "one update" is portable reasoning; promising exact SQL text is not.

Flush is not commit

Flush synchronizes persistence-context changes with the database transaction. Commit completes the transaction. Therefore:

TEXT

```
Java mutation -> flush executes SQL -> later error -> rollback
```

SQL happened, but no durable commit remained. Calling `flush()` is useful to detect database failures earlier or satisfy query consistency; it does not make a transaction durable.

With JPA `AUTO`, a query in a joined transaction must see relevant pending changes, so the provider may flush before it. With `COMMIT`, visibility of unflushed changes to queries is less predictable under the portable contract. Outside an active joined transaction, the provider must not flush.

persist, merge, and the ignored-return trap

`persist(newEntity)` makes that instance managed. `merge(detached)` copies state into a managed instance and returns it; the original remains detached.

JAVA

```
PurchaseOrder managed = entityManager.merge(detached);  
// continue with managed, not detached
```

Blindly merging a whole client graph can overwrite concurrent changes or cascade unexpected inserts. Load the aggregate, authorize/validate the command, and mutate explicit fields.

Clear, detach, and refresh

- `detach(entity)`: stop tracking one entity.
- `clear()`: detach all managed entities; useful in large batches.
- `refresh(entity)`: overwrite managed state from the database; pending local changes can be lost.

Flush before clear when work must reach the transaction. Batch loops often `flush()` and `clear()` every chunk to bound memory.

Failure timeline

TEXT

```
persist duplicate business key
-> Java call may return
-> unrelated query triggers AUTO flush
-> unique constraint fails at the query line
```

The query did not create the duplicate; it exposed deferred SQL. Diagnose from transaction history, not merely the top stack-frame line.

Quick check and practice

- 1 Is the object passed to `merge` made managed?
 - 2 Why can a select throw an insert constraint exception?
 - 3 What survives a flush followed by rollback?
 - 4 When should a batch clear its context?
- **Foundation:** Label six objects as new, managed, detached, or removed.
 - **Interview Core:** Predict SQL for find-mutate-query-commit under `AUTO`.
 - **SDE-2 Follow-up:** Repair a 500,000-row batch whose persistence context grows until OOM.

SDE-2 CORE CHAPTER

Chapter 5 - Associations, Ownership, Cascades, Orphans, and Aggregate Boundaries

Associations are where convenient object navigation most often creates incorrect SQL and uncontrolled graphs.

Foreign-key ownership

For an order with lines, the foreign key is on `order_line`. The child side usually owns the relationship:

JAVA

```
@Entity
class OrderLine {
    @ManyToOne(fetch = FetchType.LAZY, optional = false)
    @JoinColumn(name = "order_id", nullable = false)
    private PurchaseOrder order;
}

@Entity
class PurchaseOrder {
    @OneToMany(mappedBy = "order",
        cascade = CascadeType.ALL,
        orphanRemoval = true)
    private List<OrderLine> lines = new ArrayList<>();

    public void addLine(OrderLine line) {
        lines.add(line);
        line.attachTo(this);
    }

    public void removeLine(OrderLine line) {
        lines.remove(line);
        line.detachFrom(this);
    }
}
```

`mappedBy` says the collection is inverse; changing only it may leave the owning foreign-key side unchanged. Helper methods keep both in-memory sides consistent.

Cascade is an entity-operation policy

Cascade tells JPA whether operations such as persist, merge, remove, refresh, or detach propagate across an association. It is not the same as database `ON DELETE CASCADE`, which the database enforces even for direct SQL.

Use cascade from an aggregate root to privately owned children. Avoid `CascadeType.REMOVE` across shared many-to-many relationships: deleting one user must not delete shared roles.

Orphan removal

With `orphanRemoval = true`, removing a child from the owned association schedules deletion of that child row. This fits order lines that have no meaning outside the order. It is not a generic "clean unused rows" feature.

Expected trace:

SQL

```
select ... from purchase_order where order_id=?;
select ... from order_line where order_id=?;
delete from order_line where line_id=?;
```

The delete is normally emitted at flush, not at `List.remove` itself.

Aggregate boundaries

An aggregate is a consistency boundary, not every object reachable by a getter. Keep transactional invariants local. References to another aggregate often need only its identifier or a lazy many-to-one, not cascading lifecycle ownership.

Large bidirectional graphs increase:

- accidental loads and serialization recursion;
- cascade traversal and dirty-check work;
- unclear delete responsibility;
- equality/debugging complexity.

Prefer unidirectional mappings unless both directions serve real domain operations.

Many-to-many warning

A direct `@ManyToMany` hides the join table as soon as the relationship needs attributes such as role, created time, ordering, or status. Model an explicit association entity (`Membership`) with its own identity/invariants. It also gives precise update/delete control.

Collection choices

- `List` permits order/duplicates; persistent ordering needs `@OrderColumn` or an ordered query/mapping.
- `Set` depends on stable `equals/hashCode`.
- `Map` requires deliberate persistent key semantics.

Do not choose `Set` as a performance hack. Database uniqueness still needs a constraint.

Failure matrix

Symptom	Cause	Repair
child FK remains null	only inverse side changed	synchronize owning side
deleting group deletes users	remove cascade across shared relation	model ownership correctly
removed child row remains	no orphan removal/direct FK update	define lifecycle and inspect flush SQL
stack overflow in JSON/logging	bidirectional traversal	DTO boundary and controlled <code>toString</code>
duplicate children despite <code>Set</code>	unstable equality/no DB constraint	stable equality plus unique constraint

Practice

- **Foundation:** Identify the owner for a foreign key on `order_line`.
- **Interview Core:** Choose cascade and orphan behavior for order lines versus customer.
- **SDE-2 Follow-up:** Replace a many-to-many user-role mapping when membership gains expiration and audit fields.

SDE-2 CORE CHAPTER

Chapter 6 - Proxies, Fetch Plans, N+1, and Graph Control

Fetching is a use-case decision: which state must this transaction load, in how many statements, with what row multiplication and memory cost?

Lazy does not mean "never loaded"

A lazy association is represented by a proxy or persistent collection wrapper and initialized when accessed while a usable persistence context exists.

TEXT

```
load order -> proxy customer / uninitialized lines
access order.getLines().size()
    -> SQL if context is open
    -> LazyInitializationException if detached and not initialized
```

Do not repair a detached lazy failure by making everything eager or keeping a persistence context open through web rendering. Define a transactional query that returns the data the application use case needs.

The N+1 shape

JAVA

```
List<PurchaseOrder> orders = entityManager
    .createQuery("select o from PurchaseOrder o", PurchaseOrder.class)
    .getResultList(); // 1 query

for (PurchaseOrder order : orders) {
    System.out.println(order.getLines().size()); // up to N queries
}
```

Trace:

SQL

```
select ... from purchase_order;           -- 1
select ... from order_line where order_id=?; -- repeated per order
```

The code has no explicit query in the loop, which is exactly why query-count tests and SQL traces matter.

Fetch join

JPQL

```
select distinct o
from PurchaseOrder o
left join fetch o.lines
where o.id = :id
```

This is effective for one aggregate or a bounded set. Collection fetch joins multiply result rows. Fetching two bag-like collections can create a Cartesian explosion or provider limitation. `distinct` can deduplicate root entities in Java/SQL semantics; it does not make transferred child combinations free.

Entity graphs

An entity graph describes attributes to fetch without embedding fetch joins into every JPQL statement. It improves use-case-level plans, but still inspect generated SQL and provider semantics. A graph is not a guarantee of one query in every provider/mapping combination.

Batch and subselect fetching

Hibernate batch fetching can initialize several lazy proxies/collections with `IN ( . . . )` queries, reducing N+1 without joining a wide Cartesian result. Subselect fetching can load related collections for roots from a prior query. These are provider-specific tuning options: measure query count, parameter limits, row count, and memory.

Pagination and collection fetch joins

Applying offset/limit to a collection fetch join is dangerous: SQL rows represent root-child pairs, not distinct roots. Providers may paginate in memory or reject/warn, producing incorrect performance assumptions.

Use two steps:

- 1 page only root IDs with deterministic order;
- 2 fetch roots and required children for those IDs;
- 3 restore requested order if the second query does not preserve it.

Or use projections when the response does not require entities.

DTO boundary

Serializing entities directly can trigger lazy SQL, recursion, expose internal columns, and accept unsafe detached updates. Map to a response DTO inside the transaction with an explicit fetch plan or query projection.

Fetch decision table

Use case	Technique	Risk to test
one aggregate detail	fetch join/entity graph	row multiplication
page of summaries	DTO projection	constructor/alias drift
page roots then children	two-step ID page + fetch	ordering and duplicate IDs
many roots, occasional child access	lazy + batch fetching	query count/parameter size
detached API response	assemble DTO in transaction	accidentally accessed field

Interview drill

Question: "Why not set every association to lazy?"

Answer: Lazy is a safe default for avoiding unconditional work, not a complete fetch plan. Each use case still needs explicit loading. Otherwise access patterns create N+1 or detached failures.

Practice

- **Foundation:** Predict query count for ten orders whose lines are accessed.
- **Interview Core:** Repair it with both a fetch join and projection; compare trade-offs.
- **SDE-2 Follow-up:** Page customers with two collections without Cartesian multiplication.

SDE-2 CORE CHAPTER

Chapter 7 - JPQL, Criteria, Native SQL, Projections, and Pagination

JPQL queries the entity model

JAVA

```
List<OrderSummary> summaries = entityManager.createQuery("""
    select new interview.OrderSummary(o.id, o.totalCents, o.createdAt)
    from PurchaseOrder o
    where o.customerId = :customerId
        and o.status = :status
    order by o.createdAt desc, o.id desc
""", OrderSummary.class)
    .setParameter("customerId", customerId)
    .setParameter("status", OrderStatus.PAID)
    .setMaxResults(50)
    .getResultList();
```

JPQL names entities and attributes, not tables and columns. Constructor projections return unmanaged DTOs and avoid tracking a graph that will not be updated.

Joins and implicit navigation

Explicit joins make cardinality and join type visible. Navigating `o.customer.email` can create an implicit inner join and eliminate rows where the relationship is absent. Use explicit `left join` when null-preserving behavior is required.

For collection membership and existence, express the domain question instead of fetching a collection and filtering in Java.

Criteria API

Criteria is useful for dynamic, type-oriented query composition:

JAVA

```
CriteriaBuilder cb = entityManager.getCriteriaBuilder();
CriteriaQuery<PurchaseOrder> query = cb.createQuery(PurchaseOrder.class);
Root<PurchaseOrder> order = query.from(PurchaseOrder.class);

List<Predicate> predicates = new ArrayList<>();
if (status != null) {
    predicates.add(cb.equal(order.get("status"), status));
}
query.where(predicates.toArray(Predicate[]::new));
```

It can become verbose and stringly typed without a metamodel. Keep query composition isolated and test its generated SQL/results. Do not choose Criteria merely because it looks more "enterprise."

Native SQL

Use native SQL for database-specific features, window/CTE/reporting shapes not expressed cleanly, optimizer hints only when evidence demands them, or carefully tuned hot paths.

Trade-offs include dialect coupling, manual result mapping, persistence-context staleness after writes, and migration coordination. Parameterize values; native does not mean concatenate.

Projections

- **Entity:** managed identity and changes; load only when you need domain behavior.
- **Scalar/tuple:** compact but alias/index handling can be fragile.
- **DTO constructor:** explicit read contract and no dirty checking.
- **Interface/record via framework mapping:** convenient but implementation rules vary.

Selecting an entity plus one scalar still returns/maintains the entity; it is not automatically a lightweight projection.

Pagination

JPA offset pagination uses `setFirstResult` and `setMaxResults`; it inherits deep-offset cost and concurrent-shift behavior.

Keyset JPQL for descending (`createdAt`, `id`):

JPQL

```
where o.createdAt < :time
    or (o.createdAt = :time and o.id < :id)
order by o.createdAt desc, o.id desc
```

Use a complete stable ordering key, fetch one extra row, and return an opaque cursor. Seek support from provider-specific APIs may reduce boilerplate, but the relational predicate remains the contract.

Count-query trap

A page often requires a separate count. A fetch join, `distinct`, grouping, or complex predicate can make automatically derived counts slow or wrong. A `Slice`/cursor response avoids total count when the product needs only "has next." If total count is required, write and test its query intentionally.

Bulk DML bypasses managed state

JAVA

```
int changed = entityManager.createQuery("""
    update PurchaseOrder o
    set o.status = :expired
    where o.status = :created and o.createdAt < :cutoff
    """)
    .setParameter("expired", EXPIRED)
    .setParameter("created", CREATED)
    .setParameter("cutoff", cutoff)
    .executeUpdate();
```

Bulk JPQL does not synchronize each managed entity or invoke ordinary entity lifecycle/version handling in the same way. Flush relevant pending changes first, then clear/refresh the context, and decide how version/concurrency should be preserved. Directly adding `version = version + 1` may be required by policy, but affected stale instances still need reconciliation.

Practice

- **Foundation:** Convert one SQL filter into JPQL and name the generated joins.
- **Interview Core:** Build a projection and keyset continuation query.
- **SDE-2 Follow-up:** Design a bulk expiry job that does not leave managed or cached state stale.

SDE-2 CORE CHAPTER

Chapter 8 - Batching, Locking, Caches, Equality, and Bulk Work

JDBC batching through Hibernate

Configure a bounded `hibernate.jdbc.batch_size`, order compatible inserts/updates when appropriate, and flush/clear in chunks:

JAVA

```
for (int index = 0; index < commands.size(); index++) {
    entityManager.persist(toEntity(commands.get(index)));
    if ((index + 1) % batchSize == 0) {
        entityManager.flush();
        entityManager.clear();
    }
}
```

Batching can be limited by identity ID generation, mixed statement shapes, early flushes, generated-key needs, or driver settings. Hibernate statistics showing insert count does not prove network batches; use `datasource/driver` tracing or an integration proxy when batch transport matters.

Optimistic locking

JAVA

```
@Version
private long version;
```

Hibernate reads the version and generates a predicate like:

SQL

```
update purchase_order
set status=?, version=?
where order_id=? and version=?
```

Zero updated rows becomes an optimistic-lock failure and the transaction rolls back. Retry only if you can replay the command against fresh state without duplicating external effects. Often returning conflict to the caller is more honest than silently overwriting.

Every concurrently changed entity that needs protection should be versioned. Versioning only the root does not automatically detect direct changes to every child unless the mapping/update changes the root version as intended.

Pessimistic locking

JAVA

```
PurchaseOrder order = entityManager.find(
    PurchaseOrder.class, id, LockModeType.PESSIMISTIC_WRITE);
```

JPA requires a transaction for pessimistic lock modes. The provider translates to database locks, but exact SQL, scope, timeout, and range behavior are database-specific. Keep transactions short, lock in stable order, and handle deadlocks/timeouts.

First- and second-level caches

- First-level persistence-context identity is mandatory and transaction/context local.
- Hibernate second-level cache is optional and shared by factory scope for configured entity/collection data.
- Query cache stores result identities/scalars and relies on invalidation metadata; it is not an automatic performance switch.

Cache only data with a clear freshness, invalidation, memory, and multi-node strategy. Database bulk/native changes can bypass assumptions. Measure hit rate, miss cost, invalidations, stale tolerance, and tail latency.

Equality revisited

Never generate Lombok-style equality over every entity field and association. It can initialize lazy graphs, recurse, and change hash codes. Keep entities out of hash collections until identity semantics are stable, or use an immutable natural key with database uniqueness and a proxy-compatible implementation.

Bulk versus entity-oriented work

Entity loops provide lifecycle callbacks, cascades, dirty checking, and version checks but can be memory/statement heavy. Bulk SQL/JPQL is efficient for set-based changes but bypasses per-entity behavior and tracked state. Choose deliberately:

Requirement	Better starting point
change one aggregate with invariants	managed entity
expire 5 million independent rows	bounded bulk SQL/JPQL
callback/outbox per row	set-based update plus set-based outbox design, or bounded entity chunks
database-specific upsert	native SQL/JDBC with explicit contract

Failure matrix

Failure	Cause	Response
batch still sends single statements	ID strategy/driver/shape	instrument actual JDBC transport
stale update silently wins	no @Version or bulk bypass	version/conditional SQL
pessimistic endpoint times out	long lock scope or missing index	inspect lock/query evidence
cache returns stale entity	invalidation/bulk/external writer	explicit cache policy/eviction
entity disappears from HashSet	mutable hash field	stable equality

Practice

- **Foundation:** Explain what @Version adds to UPDATE.
- **Interview Core:** Choose optimistic or pessimistic control for scarce-ticket checkout.
- **Interview Core:** Prove whether 10,000 inserts are actually batched.
- **SDE-2 Follow-up:** Safely expire sessions with bulk DML, cache eviction, and observability.

SDE-2 CORE CHAPTER

Chapter 9 - Testing, SQL Traces, and Production Diagnostics

ORM tests must prove database behavior, not only Java method calls.

Test layers

- 1 **Domain unit tests:** invariants without JPA.
- 2 **Mapping/behavior tests:** real provider plus a disposable database; lifecycle, SQL count, cascade, locking.
- 3 **Target-engine integration tests:** exact MySQL version for collation, generated DDL, plans, isolation, locks, and dialect features.
- 4 **Service tests:** transaction boundaries, exception translation, outbox/idempotency.
- 5 **Production telemetry:** query digests, traces, pool waits, locks, database metrics.

H2 is fast and portable for JPA mechanics, but compatibility mode does not turn it into MySQL. A passing H2 test cannot certify InnoDB next-key locks, MySQL collation, optimizer plans, or online DDL.

Assert outcomes and SQL budgets

For an order-detail use case, assert:

- returned state;
- transaction commit/rollback;
- constraint and version conflicts;
- bounded statement count;
- no SQL after leaving the service transaction;
- generated query shape where regression risk is high.

Avoid brittle snapshots of every generated alias. Count statements and assert important joins/predicates through datasource instrumentation or Hibernate statistics; use exact SQL snapshots selectively.

Statistics

Hibernate statistics can expose entity loads, fetches, query executions, prepared statements, cache behavior, and flushes. Enabling them has overhead; use deliberately in tests and controlled diagnostics.

JAVA

```
SessionFactory sessionFactory = entityManagerFactory.unwrap(SessionFactory.class);
Statistics statistics = sessionFactory.getStatistics();
statistics.clear();
// execute use case
long statements = statistics.getPrepareStatementCount();
```

A diagnostic sequence

When an endpoint regresses:

- 1 separate pool wait, transaction time, query time, and response mapping;
- 2 correlate trace span to normalized SQL/query count;
- 3 identify unexpected flushes or lazy loads;
- 4 inspect actual database plan and lock waits;
- 5 compare bind/tenant distribution;
- 6 reproduce with production-like cardinality;
- 7 fix query/fetch/index/transaction scope and add a regression test.

Symptom-to-cause map

Symptom	Likely ORM cause	Evidence
endpoint query count grows with rows	N+1	statement count and stack trace per query
exception occurs on unrelated select	AUTO flush exposed deferred failure	transaction log and flush event
update missing after bulk DML	stale managed object overwrote/ignored DB state	clear/refresh and SQL timeline
memory grows in import	persistence context retains entities	managed count/heap and absent clear
lock held during HTTP call	oversized transaction boundary	transaction trace and DB lock duration
page duplicates/misses	unstable sort or collection join pagination	SQL rows and ordering key

Test isolation

Rollback-after-each-test is convenient but can hide commit-time behavior. Add tests that truly commit for constraints, optimistic locks, outbox state, and callbacks. Generate unique keys, avoid order-dependent fixtures, and reset caches/statistics between tests.

Security and logging

SQL logs can expose tokens, emails, and personal data. Use redaction and controlled sampling. Do not enable verbose binds globally during an incident without scope, retention, and access controls.

Practice

- **Foundation:** Write a test proving dirty checking updates at commit.
- **Interview Core:** Add a query-count test that fails when N+1 returns.
- **Interview Core:** Test stale optimistic updates with two persistence contexts.
- **SDE-2 Follow-up:** Build a target-MySQL test for a lock-range claim while keeping H2 tests fast.

SDE-2 CORE CHAPTER

Chapter 10 - Hibernate/JPA Live Interviews, Answered Questions, and Readiness

Live interview 1: N+1 in production

Interviewer: "A list endpoint went from 20 to 2,001 SQL statements. Fix it."

Candidate: "I would confirm query count and which getter initializes the association. For summary rows I prefer a DTO projection. If a bounded aggregate is needed, use a fetch join/entity graph; if paging roots with collections, page root IDs then fetch children. I would compare rows transferred, not merely query count, and add a statement-budget regression test."

Live interview 2: save() and transaction truth

Interviewer: "The code called save, then a remote API, then commit failed. What happened?"

Candidate: "For a managed/new entity, save/persist may only schedule state. SQL or a deferred constraint can fail at flush/commit after the external call. A database transaction cannot roll back that remote effect. I would persist state plus an outbox atomically, commit, then perform the remote action idempotently."

Live interview 3: detached update

Interviewer: "The controller posts an entity JSON and we call `merge`. Any issue?"

Candidate: "It permits over-posting and copies detached graph state, potentially overwriting fields the client never owned and concurrent changes. I would accept a command DTO, load the aggregate in a transaction, authorize and validate explicit mutations, and rely on `@Version` for conflict detection."

Live interview 4: optimistic versus pessimistic

Interviewer: "Which lock for the final concert ticket?"

Candidate: "I first ask collision rate, invariant, transaction duration, and failure cost. A single conditional SQL update can be best. Optimistic versioning suits low contention and replayable commands; pessimistic locking avoids late conflict but holds scarce resources and can deadlock. I would load test the chosen strategy and keep external calls outside the lock."

Live interview 5: collection fetch plus paging

Interviewer: "Why does `join fetch o.lines` break our page?"

Candidate: "The SQL page is over root-child rows, so one root consumes several rows and the provider may paginate in memory. I would page ordered root IDs, then fetch roots and lines in a second query, or return a projection. The count query is separate and must not inherit the collection join."

Live interview 6: bulk update bug

Interviewer: "A JPQL bulk update changed rows, but the response returned old values."

Candidate: "Bulk DML bypasses the managed entity state. Flush relevant pending work before it, execute the bulk statement, then clear or refresh. I would also define version/cache invalidation behavior and keep the bulk operation in its own explicit boundary."

Live interview 7: batch import OOM

Interviewer: "Hibernate import crashes after 300,000 rows."

Candidate: "The persistence context likely retains every managed entity, and statements may not batch due to identity generation or mixed shapes. I would use bounded transactions/chunks, flush and clear each chunk, configure and instrument JDBC batching, and consider set-based JDBC/native loading. I would make the job restartable and idempotent."

Rapid answered questions

- 1 **JPA versus Hibernate?** JPA is the portable contract; Hibernate is a provider and extension set.
- 2 **Is EntityManager thread-safe?** No; scope it to a unit of work/request transaction.
- 3 **Does persist always insert immediately?** No; SQL timing depends on flush and ID strategy.
- 4 **Does flush commit?** No; it synchronizes SQL inside the current transaction.
- 5 **What does merge return?** A managed instance containing copied state; the argument remains detached.
- 6 **What is the first-level cache?** The persistence context's identity map, not a cross-request cache.
- 7 **Owning side?** The side whose mapping controls the foreign key/join table update.
- 8 **Cascade versus orphan removal?** Cascade propagates entity operations; orphan removal deletes a privately owned child removed from its association.
- 9 **Cascade versus ON DELETE CASCADE?** ORM operation propagation versus database referential action.
- 10 **Lazy loading failure outside transaction?** The proxy/collection lacks an open usable context; fetch a DTO/graph inside the service boundary.
- 11 **Why not eager everywhere?** It causes unconditional joins/secondary selects and graph explosion.
- 12 **What is N+1?** One root query followed by a query per result/association access.
- 13 **Can a fetch join always fix N+1?** No; multiple collections and paging can multiply rows or break pagination.
- 14 **JPQL names what?** Entity types and attributes, not physical table/column names.
- 15 **What does @Version do?** Adds version verification/update so stale writes affect zero rows and fail optimistically.
- 16 **Does bulk JPQL honor each entity callback/version?** It bypasses ordinary per-entity lifecycle/tracked state; manage version and context explicitly.
- 17 **Second-level cache by default?** Provider/configuration dependent and optional; first-level context is fundamental.
- 18 **Why can equality load SQL?** Generated equality may traverse lazy associations/proxies.
- 19 **Why can a query trigger an insert error?** AUTO flush synchronizes pending inserts before a relevant query.
- 20 **Can H2 prove MySQL locking behavior?** No; use the exact engine/version for dialect, plan, isolation, and locks.

Cumulative assessment

Implement and explain an order aggregate with lines:

- 1 schema and expected SQL first;
- 2 IDs, value mappings, version, and constraints;
- 3 ownership, cascade, orphan behavior, and helper methods;
- 4 detail and summary fetch plans with query-count budgets;
- 5 optimistic concurrency conflict test;
- 6 bulk expiry job with clear/cache policy;
- 7 batch import with bounded context;
- 8 target-MySQL tests for locks and plans;
- 9 incident runbook for N+1, pool wait, and slow flush.

Readiness rubric

Level	Evidence
Foundation	maps entities and explains lifecycle/transactions
Interview Core	predicts SQL, owns associations, controls fetch and versions
SDE-2 ready	reasons across ORM, JDBC, database, failure, and observability

Selected authoritative references

- Jakarta Persistence 3.2 specification: entity lifecycle, persistence context, flush, query, locking, and cache contracts.
- Hibernate ORM User Guide (current release): provider-specific mappings, fetching, batching, caches, statistics, and SQL behavior.
- MySQL 8.4 Reference Manual and Connector/J guide: database isolation, plans, dialect behavior, and JDBC driver settings.

Version-label provider-specific behavior and verify it against the runtime in the lab or target service. The accompanying H2 fixture validates portable ORM mechanics, not MySQL-specific plans, collations, or locks.

SDE-2 CORE CHAPTER

Chapter 11 - Java 21 Hibernate and JPA Interview Companion

Executable models for entity lifecycle, persistence-context identity, dirty checking, association ownership, optimistic versions, and bounded batch work.

JAVA (continued 1/4)

```
import java.util.ArrayList;
import java.util.HashMap;
import java.util.List;
import java.util.Map;
import java.util.Objects;

/** Dependency-free executable models for the Hibernate/JPA interview volume. */
public final class HibernateJpaInterviewCompanion {
    private HibernateJpaInterviewCompanion() {}

    enum Lifecycle {
        NEW, MANAGED, DETACHED, REMOVED
    }

    static Lifecycle persist(Lifecycle state) {
        if (state != Lifecycle.NEW) {
            throw new IllegalStateException("persist expects a new instance");
        }
        return Lifecycle.MANAGED;
    }

    static Lifecycle detach(Lifecycle state) {
        return state == Lifecycle.MANAGED || state == Lifecycle.REMOVED
            ? Lifecycle.DETACHED : state;
    }

    record EntityKey(Class<?> type, long id) {
        EntityKey {
            Objects.requireNonNull(type, "type");
        }
    }
}
```

JAVA (continued 2/4)

```
static final class IdentityMap {
    private final Map<EntityKey, Object> managed = new HashMap<>();

    <T> T manage(EntityKey key, T entity) {
        Object existing = managed.putIfAbsent(key, entity);
        @SuppressWarnings("unchecked")
        T canonical = (T) (existing == null ? entity : existing);
        return canonical;
    }

    void clear() {
        managed.clear();
    }
}

record Snapshot(String status, long totalCents) {}

static boolean isDirty(Snapshot loaded, Snapshot current) {
    return !Objects.equals(loaded, current);
}

static final class Order {
    private final long id;
    private final List<Line> lines = new ArrayList<>();

    Order(long id) {
        this.id = id;
    }

    void add(Line line) {
        Objects.requireNonNull(line, "line");
        lines.add(line);
    }
}
```


JAVA (continued 3/4)

```

        line.order = this;
    }

    void remove(Line line) {
        if (lines.remove(line)) {
            line.order = null;
        }
    }
}

static final class Line {
    private Order order;
}

record VersionedState(String status, long version) {
    VersionedState update(String nextStatus, long expectedVersion) {
        return version == expectedVersion
            ? new VersionedState(Objects.requireNonNull(nextStatus), version + 1)
            : this;
    }
}

static <T> List<List<T>> batches(List<T> input, int batchSize) {
    if (batchSize <= 0) {
        throw new IllegalArgumentException("batchSize must be positive");
    }
    List<List<T>> result = new ArrayList<>();
    for (int start = 0; start < input.size(); start += batchSize) {
        result.add(List.copyOf(input.subList(start, Math.min(start + batchSize, input.size()))));
    }
    return List.copyOf(result);
}

```

JAVA (continued 4/4)

```

public static void main(String[] args) {
    assert persist(Lifecycle.NEW) == Lifecycle.MANAGED;
    assert detach(Lifecycle.MANAGED) == Lifecycle.DETACHED;

    IdentityMap context = new IdentityMap();
    EntityKey key = new EntityKey(Order.class, 7);
    Order first = context.manage(key, new Order(7));
    Order second = context.manage(key, new Order(7));
    assert first == second;

    assert !isDirty(new Snapshot("CREATED", 500), new Snapshot("CREATED", 500));
    assert isDirty(new Snapshot("CREATED", 500), new Snapshot("PAID", 500));

    Line line = new Line();
    first.add(line);
    assert line.order == first && first.lines.size() == 1;
    first.remove(line);
    assert line.order == null && first.lines.isEmpty();

    VersionedState state = new VersionedState("CREATED", 2);
    assert state.update("PAID", 1).equals(state);
    assert state.update("PAID", 2).version() == 3;

    assert batches(List.of(1, 2, 3, 4, 5), 2).equals(
        List.of(List.of(1, 2), List.of(3, 4), List.of(5)));

    context.clear();
    System.out.println("HibernateJpaInterviewCompanion checks passed");
}
}

```

Part II - Practice and Handoff

TRY IT | Practice Ladder and Next Steps

TRY IT | Practice ladder

- Foundation: entity mapping and lifecycle
- Interview Core: fetching, queries, and transactions
- SDE-2 Follow-up: locks, batching, and diagnosis

For every coding problem, state the input contract, select the numeric and collection types deliberately, write the invariant before the loop or recursion, test the smallest and largest valid inputs, and close with time and auxiliary-space complexity.

Navigation

[Previous book: FW 01 - MySQL for Java Backend Interviews](#)

[Series index](#)

[Next book: FW 03 - Spring Framework for Java Engineers](#)

TRY IT | Completion check

- ☐ I can recognize the core patterns without being told the data structure or algorithm.
- ☐ I can implement the representative Java templates from memory.
- ☐ I can explain why each implementation is correct.
- ☐ I can test boundaries, invalid inputs, overflow, and adversarial shapes.
- ☐ I can answer at least one SDE-2 follow-up involving trade-offs or production constraints.

Series Roadmap

Choose Java, DSA, Frameworks, or System Design first, then follow the books inside that segment in order. The same series codes and positions appear on the website, PDF covers, and canonical download folders.

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Java	JAVA 02	Git and GitHub for Java Engineers
Java	JAVA 03	Maven and Gradle for Java Engineers
Java	JAVA 04	Language, OOP, and Modern Java
Java	JAVA 05	Collections, Streams, and I/O
Java	JAVA 06	JVM and Java Execution
Java	JAVA 07	Java Concurrency and the Memory Model
Java	JAVA 08	Performance, Diagnostics, and GC Incidents
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SEGMENT	BOOK	FOCUSED LEARNING PATH
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Frameworks	FW 07	MongoDB for Java Backend Interviews
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About the Author

Vinay Reddy Kalluri

Vinay Reddy Kalluri is a senior Java backend engineer and the author and editor of the Java SDE-2 Interview Preparation Series. His work spans high-throughput microservices, event-driven architecture, resilient APIs, large-scale data processing, cloud delivery, and production performance across healthcare and enterprise systems.

Editorial approach

Vinay sets the learning sequence and publication standard and performs the final review for Java accuracy, evidence, attribution, and release readiness. Individual contributors retain visible credit for accepted original work through AUTHORS.md and the public Git history.

What distinguishes his approach is the combination of hands-on system design and operational ownership. He treats timeouts, retries, idempotency, dead-letter recovery, data consistency, SQL behavior, caching, and observability as part of the design rather than afterthoughts. He has also mentored engineers and led modernization, migration, and reliability work across distributed services.

Selected engineering and academic highlights

- More than eight years building Java and Spring Boot services for high-throughput healthcare and enterprise platforms.
- Designed Kafka-based event systems that have processed more than one million events per hour, with explicit idempotency, retry, recovery, and observability controls.
- M.S. in Computer Science from the University of Missouri-Kansas City (3.9/4.0) and M.S. in Software Engineering from VIT University (9.3/10), where he was a Gold Medalist.
- Independent builder of Work Visa Insights, an immigration-data intelligence platform, and Play Together, a published Android application.

Why this series exists

Vinay created this series to turn fragmented interview preparation into a deliberate progression: learn the fundamentals, run the examples, predict behavior, debug failures, explain trade-offs, and only then move into advanced engineering. The books reflect the same principle he applies to production systems: clarity before cleverness, explicit contracts, measurable behavior, and reliable foundations.

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Repository: <https://github.com/vinayreddykalluri/SDE2-Interview-Handbook>. Attribution does not imply endorsement. Java and related marks are owned by their respective holders.

Examples target Java 21 unless a section states otherwise. Validate release-specific APIs, JVM behavior, security, performance, licensing, and operational requirements before production use.

Series position: Frameworks, Data, and Messaging - Book 02 of 12 - Study Step FW-02. The local table of contents and PDF bookmarks navigate within this file; the roadmap and printed filenames navigate across the complete series.

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